CHAPTER 5
Mole Concept, Stoichiometry and Behaviour of Gases

Chapter 5

Assessment Test I

1. The equation $V_1T_1 = V_2T_2$ implies that the volume occupied by a given mass of a gas is inversely proportional to the absolute temperature. However, according to Charles' law, the volume occupied by a given mass of a gas is directly proportional to the absolute temperature.

   Hence, the correct option is (d).

2. Assertion is a true statement as it is the definition of mole. However, 1 mole of any gas occupies 22.4 *l* volume only at STP and not at other condition. The reason is wrong.

   Hence, the correct option is (c).

3. (A) H_2SO_4 – 7 moles of atoms
   (B) Al_2O_3 – 5 moles of atoms
   (C) $K_2Cr_2O_7$ – 11 moles of atoms
   (D) $FeSO_4$ – 6 moles of atoms
   (E) NH_4HCO_3 – 10 moles of atoms
   Hence, the correct option is (d).

4. When a cooking gas cylinder is kept in hot water, due to increase in temperature, the kinetic energy of gas molecules increases. As a result, the number of collisions in the cylinder increases and hence, pressure increases. Therefore, further release of gas from the cylinder takes place till the pressure becomes equal to the external pressure.

   Hence, the correct option is (a).

5. The formula of magnesium nitride is Mg_3N_2 as Mg^{+2} is magnesium ion and N^{-3} is nitride ion. The formula of calcium phosphate is $Ca_3(PO_4)_2$ as Ca^{+2} is calcium ion and PO_4^{-3} is phosphate ion.

   Hence, the correct option is (b).

6. $V_1 = 1000$ ml, $V_2 = ?$
   Let $P_1 = 100$ mm and $P_2 = 120$ mm
   $$V_2 = \frac{P_1V_1}{P_2} = \frac{100\times1000}{120} = 833 \text{ ml}$$
   Hence, the correct option is (c).

7. $PV = nRT$
   $$V = \frac{nRT}{P}$$
   $n = 1$ mole
   $$\text{(A) } V = \frac{R\times200}{600} = \frac{R}{3}$$
   $$\text{(B) } V = \frac{R\times100}{400} = \frac{R}{4}$$
   $$\text{(C) } V = \frac{R\times100}{600} = \frac{R}{6}$$
   $$\text{(D) } V = \frac{R\times200}{400} = \frac{R}{2}$$
   The increasing order of volumes is $\frac{R}{6} < \frac{R}{4} < \frac{R}{3} < \frac{R}{2}$.
   Hence, the correct option is (a).

8. The formula of calcium bicarbonate is $Ca(HCO_3)_2$.
   Molecular mass = 162
   $$\text{\% of calcium} = \frac{40\times100}{162} = 24.69\%$$
   $$\text{\% of hydrogen} = \frac{2\times100}{162} = 1.23\%$$
   $$\text{\% of carbon} = \frac{2\times12\times100}{162} = 14.8\%$$
   $$\text{\% of oxygen} = \frac{6\times16\times100}{162} = 59.2\%$$

A $\rightarrow$ oxygen; B $\rightarrow$ calcium, C $\rightarrow$ carbon; D $\rightarrow$ hydrogen

This can also be done by comparing the molecular compositions.

Hence, the correct option is (b).

9. Mass of 5.6 l of gas at STP = x g

Mass of 22.4 l of gas at STP

$= \frac{x \times 22.4}{5.6} = 4x$ g

Molecular mass of the gas = $4x$

Vapour density of the gas = $\frac{4x}{2} = 2x$

Hence, the correct option is (d).

10. $n = \frac{\text{Molecular mass}}{\text{Empirical formula mass}} = 6$

Molecular formula = Empirical formula $\times$ 6 = $CH_2O \times 6 = C_6H_{12}O_6$

The suitable formula for the above condition could be $C_6H_{12}O_6$.

Hence, the correct option is (b).

11. Formula of trivalent metallic oxide is M_2O_3.

31.6 g of oxygen = 100 g of metallic oxide

48 g of oxygen = $\frac{100 \times 48}{31.6}$

= 152 g of M_2O_3

Amount of metal in 152 g of M_2O_3

= 152 – 48 = 104 g

Atomic weight of metal = $\frac{104}{2} = 52$

Hence, the correct option is (c).

12. Percentage of CO_2 formed = 15.7 g

Amount of CO_2 formed for 70 g of mixture

$= \frac{15.7 \times 70}{100} = 11$ g

$2NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$

2 moles of $NaHCO_3 \rightarrow$ 1 mole CO_2

44 g of CO_2 is obtained from 2 $\times$ 84 g $NaHCO_3$

11 g of CO_2 is obtained from $\frac{2 \times 84 \times 11}{44} = 42$ g $NaHCO_3$

Amount of $NaHCO_3$ in the mixture = 42 g

Amount of Na_2CO_3 in the mixture = 70 – 42 = 28 g

Hence, the correct option is (c).

13. No. of moles of $KClO_3$ taken = $\frac{1225}{122.5} = 10$ moles

No. of moles of oxygen formed = $\frac{224}{22.4} = 10$ moles

$2KClO_3 \rightarrow 2KCl + 3O_2$

10 moles of 100% pure $KClO_3 \rightarrow$ 15 moles of oxygen

15 moles of oxygen = 10 moles $KClO_3$

10 moles of oxygen = $\frac{100}{15}$

= 6.6 moles $KClO_3$

$\therefore$ The percentage of pure $KClO_3$ in the sample = 66%

Hence, the correct option is (c).

14. (i) $\rightarrow$ (b)

$C_2H_2 + 5/2O_2 \rightarrow 2CO_2 + H_2O$

(ii) $\rightarrow$ (d)

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

(iii) $\rightarrow$ (a)

$C_2H_4 \rightarrow 3O_2 \rightarrow 2CO_2 + 2H_2O$

(iv) $\rightarrow$ (c)

$C_2H_6 + 7/2O_2 \rightarrow 2CO_2 + 3H_2O$

Hence, the correct option is (a).

15. Balanced chemical equation for the complete combustion of one mole a hydrocarbon is

Given that

$\frac{y}{2} = x + 1$ [From (1)]

$y = 2x + 2$

Given that

$x + \frac{y}{4} = \frac{13}{2}$ [From (2)]

$x + \frac{2x+2}{4} = \frac{13}{2} \Rightarrow \frac{6x+2}{4} = \frac{13}{2}$

$12x = 52 - 4 = 48 \Rightarrow x = \frac{48}{12} = 4$

$\therefore \frac{y}{2} = x + 1 = 4 + 1 = 5 \Rightarrow \therefore y = 10$

$x = 4, y = 10, C_xH_y = C_4H_{10}$.

Hence, the correct option is (a).

Assessment Test II

1. The given graph shows that the variable taken on Y axis remains constant with variation in temperature.
   As $V \alpha T$, $\frac{V}{T}$ is constant at constant pressure.
   Therefore, a graph of $\frac{V}{T}$ Vs. T runs parallel to the temperature axis.
   Hence, the correct option is (c).

2. Since 1 mole of any gas occupies 22.4 l volume at STP, 2 moles of the same gas should occupy 44.8 l volume at STP. Volume of a given mass of a gas necessarily depends on the temperature and pressure.

   Hence, the correct option is (a).

3. (i) 2 moles of $H_2O_2 \rightarrow 4$ N oxygen atoms
   (ii) 1 moles of $Fe_2O_3 \rightarrow 3$ N oxygen atoms
   (iii) 0.5 moles of $H_2SO_4 \rightarrow 2$ N oxygen atoms
   (iv) 1 mole of $H_2O \rightarrow 1$ N oxygen atoms
   (v) 0.25 moles of $HNO_3 \rightarrow 0.75$ N oxygen atoms.
   ∴ DEABC is the required order.
   Hence, the correct option is (c).

4. A balloon filled with air and tightly tied, when hung close to the source of heat, bursts after some time as the pressure inside the balloon increases with increase in temperature.

   Hence, the correct option is (c).

5. The formula M_2HPO_4 corresponds to M^{+1} and HPO_4^{-2} radicals. M^+ ion can form $MClO_4$, M_2CO_3 and MH with the radicals ClO^-, CO_3^{-2}, and H^-, respectively.

   Hence, the correct option is (c).

6. $V_1 = 6\, l$
   $V_2 = 8\, l$
   Let $P_1 = 100$ mm of Hg
   $P_2 = ?$
   $$P_2 = \frac{P_1V_1}{V_2} = \frac{100 \times 6}{8} = 75 \text{ mm of Hg}$$
   25% decrease in pressure is found.
   Hence, the correct option is (d).

7. $PV = nRT \Rightarrow P = \frac{nRT}{V} = \frac{RT}{V}$ (n = constant)

   (a) $P = \frac{R \times 200}{200} = R$

   (b) $P = \frac{R \times 300}{600} = \frac{R}{2}$

   (c) $P = \frac{R \times 400}{200} = 2R$

   (d) $P = \frac{R \times 500}{600} = \frac{5R}{6}$

   The correct decreasing order of pressure exerted by the gases are $2R > \frac{5R}{6} > R > \frac{R}{2}$.
   Hence, the correct option is (b).

8. The formula of sodium bicarbonate is $NaHCO_3$.
   Molecular mass = 84

   % of sodium $= \frac{2300}{84} = 27.4\%$

   % of hydrogen $= \frac{100}{84} = 1.2\%$

   % of carbon $= \frac{1200}{84} = 14.3\%$

   % of oxygen $= \frac{4800}{84} = 57.11\%$

   P = carbon, Q = hydrogen, R = sodium, S = oxygen
   This can be done by comparing the molecular composition.
   Hence, the correct option is (d).

9. Vapour density of gas = x

   Molecular mass of gas = $2x$
   $2x$ g of gas contains 6×10^{23} molecules.
   $\frac{3x}{2}$ g of gas contains $6 \times 10^{23} \times \frac{3X}{2} \times \frac{1}{2X}$
   $= 4.5 \times 10^{23}$ molecules
   Hence, the correct option is (a).

10. Both glucose and acetic acid have the same empirical formula, i.e., CH_2O. Neither ethyl alcohol nor sucrose has an empirical formula. The empirical formula of ethene is CH_2 and that of acetylene is CH.

    Hence, the correct option is (a).

11. 30 g oxygen – 100 g of M_2O_3

48 g oxygen $= \dfrac{4800}{30}$

$= 160$ g of M_2O_3

Mass of metal $= 160 - 48 = 112$ g

$\therefore$ GAM $= \dfrac{112}{2} = 56$ g

Hence, the correct option is (d).

12. Volume of CO_2 obtained at STP $= 5.6\ l$

No. of moles of $CO_2 = 0.25$ moles

$CaCO_3 \rightarrow CaO + CO_2$

No. of moles of $CaCO_3 = 0.25 \times 100 = 25$ g

Amount of mixture $= 50$ g

Hence, the correct option is (b).

13. $2KClO_3 \rightarrow 2KCl + 3O_2$

No. of moles of oxygen $= 134.4/22.4 = 6$ moles

6 moles of $O_2 \rightarrow 4$ moles of pure $KClO_3$

80% of x moles $= 4$

$x = \dfrac{400}{80} = 5$ moles of pure $KClO_3$

Extra amount of $KClO_3$ taken

$= 5 - 4 = 1$ mole $= 122.5$ g

Hence, the correct option is (d).

14. $C_2H_2 + 5/2O_2 \rightarrow 2CO_2 + H_2O$

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$

$C_2H_6 + 7/2O_2 \rightarrow 2CO_2 + 3H_2O$

(i) $\rightarrow$ (B); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (C)

Hence, the correct option is (b).

15. The balanced chemical equation for complete combustion of hydrocarbon is

$$C_xH_y + \left(x + \frac{y}{2}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O.$$

Given that $x + \dfrac{y}{4} = 4.5 \Rightarrow 4x + y = 18 \rightarrow$ (1)

Given that $x = \dfrac{y}{2} \Rightarrow y = 2x$

$\therefore$ From Eq. (1) $\Rightarrow 4x + 2x = 18 \Rightarrow x = 3$

$\therefore y = 2 \times 3 = 6$

$\therefore$ Hydrocarbon is C_3H_6.

Hence, the correct option is (d).

Assessment Test III

1. 44 g of CO_2 corresponds to 1 mole, which contains 6×10^{23} molecules.

1 mole of any gas occupies 22.4 L at STP.

Hence, the correct option is (d).

2. The least possible temperature is –273°C or 0 K. Any centigrade temperature greater than –273°C will become positive when converted to absolute scale. Since centigrade temperature less then –273°C is not possible, Kelvin temperature cannot have a negative value.

Hence, the correct option is (a).

3. Plumbic carbonate $= Pb^{+4}$, $CO^{-2}_3 \Rightarrow Pb(CO_3)_2$

$\Rightarrow$ 6 oxygen atoms

Ferric chromate $= Fe^{+3}$, $CrO_4^{-2} \Rightarrow Fe_2(CrO_4)_3$

$\Rightarrow$ 12 oxygen atoms

Zinc silicate $= Zn^{+2}$, $SiO_3^{-2} \Rightarrow ZnSiO_3 \Rightarrow$ 3 oxygen atoms

Aluminium sulphite $= Al^{+3}$, $SO_3^{-2} \Rightarrow Al_2(SO_3)_3$

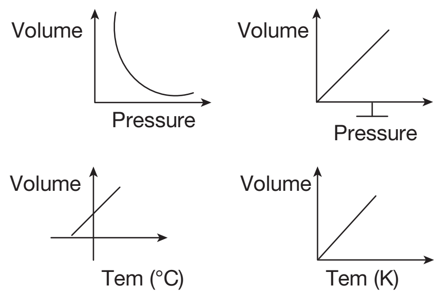
$\Rightarrow$ 9 oxygen atoms

Hence, the correct option is (b).

4. The correct order is DABC.

Hence, the correct option is (d).

5.



Hence, the correct option is (a).

6. Oxyacid is H_2SO_4.

$\therefore$ Negative radical is SO_4^{-2}.

$\therefore$ Compound of Fe^{+3}, $SO_4^{-2} \Rightarrow Fe_2(SO_4)_3$.

Hence, the correct option is (a).

7. Let $V_1 = 100\lambda$, $P_1 = 100$ atm

$V_2 = 70\lambda$, $P_2 = ?$

According to Boyle's law $P_1V_1 = P_2V_2$

$\Rightarrow P_2 = \dfrac{P_1V_1}{V_2} = \dfrac{100 \times 100}{70} = 143$ atm

$\therefore$ % change in pressure $\approx$ 43%.

Hence, the correct option is (d).

8. No. of atoms in 15×10^{25} molecules of SO_3 = $15 \times 10^{25} \times 4 = 6 \times 10^{26}$

No. of atoms in 2.5×10^{25} molecules of SO_2 = $2.5 \times 3 \times 10^{25} = 7.5 \times 10^{25}$

No. of atoms in 3×10^{25} molecules of $O_2 = 3 \times 10^{25} \times 2 = 6 \times 10^{25}$

No. of atoms in 2.5×10^{25} molecules of N_2O_5 = $2.5 \times 5 \times 10^{25} = 12.5 \times 10^{25}$

Hence, the correct option is (a).

9. $C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$

1 volume of C_2H_4 requires 3 volumes of O_2 and gives 2 volumes of CO_2.

$\Rightarrow$ 300 volume of C_2H_4 requires 900 volume of O_2 and gives 400 volume of CO_2.

Hence, the correct option is (d).

10. $2Mg(NO_3)_2 \rightarrow 2MgO + 4NO_2 + O_2$

2×148 g of $Mg(NO_3)_2$ gives 5 moles of gases ($4 \rightarrow NO_2$; $1 \rightarrow O_2$).

$\Rightarrow$ 296 g of $Mg(NO_3)_2$ gives 5×22.4 l at STP.

59.2 g of $Mg(NO_3)_2$ gives? l at STP.

$$= \frac{59.2 \times 5 \times 22.4}{296}$$

= 22.4 l at STP

Hence, the correct option is (b).

11.

Element	Atomic Weight	Percent-age	Atomic Ratio	Sim-plest Ratio
C	12	80	$\frac{18}{12} = 6.67$	$\frac{6.67}{6.67} = 1$
H	1	20	$\frac{20}{1} = 20$	$\frac{20}{6.67} = 3$

Empirical formula is CH_3.

Empirical formula mass = 15

1.5×10^{23} molecules $\rightarrow$ 7.5 g

6×10^{23} molecules $\rightarrow$?

$$= \frac{7.5 \times 6 \times 10^{23}}{1.5 \times 20^{23}} = 30 \text{ g}$$

$\therefore$ GMM = 30

$$\therefore n = \frac{30}{15} = 2$$

$\therefore$ Molecular formula = $(CH_3)_2 = C_2H_6$.

Hence, the correct option is (b).

12. $Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$

$\therefore$ Mole ratio of Fe_2O_3 and C is 1:3.

Hence, the correct option is (b).

13. $CaCO_3 \rightarrow CaO + CO_2$

100 g of $CaCO_3$ gives 22.4λ of CO_2 at STP.

x g of $CaCO_3$ gives 33.6λ of CO_2 at STP.

$$\Rightarrow x = \frac{100 \times 33.6}{22.4} = 150 \text{ g of 100\% } CaCO_3$$

$$\therefore \% \text{ purity} = \frac{150}{175} \times 100 = 85.7\%.$$

Hence, the correct option is (b).

14. 96 g of X $\rightarrow 9 \times 10^{23}$ molecules

a g of X $\rightarrow 6 \times 10^{23}$ molecules

$$\therefore a = \frac{96 \times 6 \times 10^{23}}{9 \times 10^{23}}$$

a = 64 g

$\therefore$ GMM of X = 64 g

0.32 g of Y $\rightarrow 1.2 \times 10^{22}$ molecules

b g of Y $\rightarrow 6 \times 10^{23}$ molecules

$$\Rightarrow b = \frac{0.32 \times 6 \times 10^{23}}{1.2 \times 10^{22}}$$

b = 16 g

$\therefore$ GMM of Y = 16 g

According to Graham's law $\frac{rX}{rY} = \sqrt{\frac{GMM_Y}{GMM_X}}$

$$\Rightarrow \frac{rX}{rY} = \sqrt{\frac{16}{64}} = \frac{1}{2} \Rightarrow rY = 2rX$$

Hence, the correct option is (b).

15. In an open vessel, loss in weight is due to CO_2 and water vapour.

$2\ NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$

2×84 g of $NaHCO_3 \rightarrow$ loss in weight is 44 + 18

$= 62$ g

$\Rightarrow 168 \text{ g} \rightarrow 62 \text{ g}$

210 g $NaHCO_3 = \dfrac{210 \times 62}{168} = 77.5$ g of weight loss

In a container fitted with condenser, loss of weight is due to CO_2.

$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$

168 g of $NaHCO_3 \rightarrow 44$ g of CO_2

210 g of $NaHCO_3 \rightarrow ?$

$= \dfrac{210 \times 44}{168}$

$= 55$ g of weight loss

$\therefore$ Change in weight loss $= 22.5$ g

Hence, the correct option is (b).

Assessment Test IV

1. 2 gram molecules of oxygen = 2 moles = 2 × 22.4 $\lambda = 44.8\ \lambda$ at STP

   1 mole of any gas contains 6×10^{23} particles.

   Hence, the correct option is (d).

2. Since all values of temperature are positive in Kelvin scale, the volume of gas is always positive. Hence, Kelvin temperature is used in studying gas laws.

   Hence, the correct option is (a).

3. Mercuric phosphate $= Hg^{+2}, PO_4^{-3} \Rightarrow Hg_3(PO_4)_2$ $\Rightarrow$ 13 atoms

   Magnesium nitride $= Mg^{+2}, N^{-3} \Rightarrow Mg_3N_2$ $\Rightarrow$ 5 atoms

   Cupric chlorite $= Cu^{+2}\ ClO_2^{-1} \Rightarrow Cu(ClO_2)_2$ $\Rightarrow$ 7 atoms

   Auric sulphite $= Au^{+3}, SO_3^{-2} \Rightarrow Au_2(SO_3)_2$ $\Rightarrow$ 10 atoms.

   Hence, the correct option is (c).

4. The correct order is CDAB.

   Hence, the correct option is (b).

5. Pressure – volume $\Rightarrow$ Boyle law

   Volume – number of molecules $\Rightarrow$ Avogadro law

   Volume – temperature $\Rightarrow$ Charles law

   Rate of diffusion – density $\Rightarrow$ Graham law

   Hence, the correct option is (d).

6. $CaS \Rightarrow Ca^{+2}, S^{-2}$, $Al_2(CrO_4)_3 \Rightarrow Al^{+3}, CrO_4^{-2}$

   $Al(NO_3)_3 \Rightarrow Al^{+3}, NO_3^{-1}$, $Cu(OH)_2 \Rightarrow Cu^{+2}$, OH^{-1}.

   $\therefore$ Aluminium chromate possesses trivalent cation and divalent negative radical.

   Hence, the correct option is (c).

7. $V_1 = 400\ \text{m}\lambda$, $P_2 = 3 + \left(\dfrac{3 \times 15}{100}\right) = 3.45$ atm

   $P_1 = 3$ atm, $V_2 = ?$

   According to Boyle's law, $P_1V_1 = P_2V_2$

   $\Rightarrow V_2 = \dfrac{P_1V_1}{P_2} = \dfrac{3 \times 400}{3.45} = 347\ \text{m}\lambda$

   Hence, the correct option is (b).

8. No. of moles of $O_2 = \dfrac{15 \times 10^{25}}{6 \times 10^{23}} = 0.25$ moles

   $\therefore$ Weight of $O_2 = 0.25 \times 32 = 8$ g

   Weight of $SO_3 = 0.15 \times 80 = 12$ g

   No. of moles of $SO_2 = \dfrac{2.24}{22.4} = 0.1$

   Weight of $SO_2 = 0.1 \times 64 = 6.4$ g

   No. of moles of He $= \dfrac{24 \times 10^{22}}{6 \times 10^{23}} = 0.4$

   $\therefore$ Weight of He $= 0.4 \times 4 = 1.6$ g

   $\therefore$ 24×10^{22} atoms of He possesses the least weight.

   Hence, the correct option is (d).

9. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

   1 mole of $CH_4 \rightarrow$ 2 moles of O_2.

   $C_2H_6 + 7/2O_2 \rightarrow 2CO_2 + 3H_2O$

   1 mole of $C_2H_6 \rightarrow$ 3.5 moles of O_2.

   $C_3H_6 + 9/2\ O_2 \rightarrow 3CO_2 + 3H_2O$

   1 mole of $C_3H_6 \rightarrow$ 4.5 moles of O_2.

   $C_4H_6 + 11/2\ O_2 \rightarrow 4CO_2 + 3H_2O$

   1 mole of $C_4H_6 \rightarrow$ 5.5 moles of O_2.

   Hence, the correct option is (d).

10. $C + CO_2 \rightarrow 2CO$

    12 g of C requires 22.4 λ of CO_2 at STP.

    30 g of C requires $= \dfrac{30 \times 22.4}{12}$

$= 56\lambda$ of CO_2 at STP

100% conversion requires 56 lot CO_2 at STP.

$\therefore$ 80% conversion requires ? lot of CO_2 at STP.

$\therefore V_{CO_2} = \dfrac{56 \times 80}{100} = 44.8\ \lambda$

Hence, the correct option is (a).

11. Let vapour density be x.

$\therefore$ Empirical formula mass $= x$

$\therefore n = \dfrac{2 \times \text{vapour density}}{\text{Empirical formula mass}} = \dfrac{2.x}{x} = 2$

Molecular formula $= (CH_4N)_2 = C_2H_8N_2$

The number of carbon atoms is 2.

$Na_2CO_3 \rightarrow 1$, carbon atoms; $Al_2(CO_3)_3 \rightarrow 3$

$Ca(HCO_3)_2 \rightarrow 2$ Carbon atoms

Hence, the correct option is (b).

12. $2SO_2 + O_2 \rightarrow 2SO_3$

Volume ratio of reactants = 2:1 ($\because n \ \alpha \ v$)

$H_2 + Cl_2 \rightarrow 2HCl$

Volume ratio of reactants = 1:1

$2CO + O_2 \rightarrow 2CO_2$

Volume ratio of reactants = 2:1

$N_2 + 3H_2 \rightarrow 2NH_3$

Volume ratio of reactants = 1:3

$N_2 + O_2 \rightarrow 2NO$

Volume ratio of reactants = 1:1

Hence, the correct option is (b).

13. Amounts of pure $NaHCO_3$ in 105 g sample

$= \dfrac{70 \times 105}{100} = 73.5$ g

No. of moles of $NaHCO_3 = \dfrac{73.5}{84} = 0.875$

$2NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$

1 mole of $NaHCO_3 \rightarrow 0.5$ moles of CO_2

0.875 mole of $NaHCO_3 \rightarrow 0.875 \times 0.5 = 0.437$ mole CO_2

1 mole of $CO_2 \rightarrow 22.4\ \lambda$ at STP

0.437 mole of $CO_2 \rightarrow 22.4 \times 0.437 = 9.8\ \lambda$ at STP

Hence, the correct option is (c).

14. According to Graham's law, $r \ \alpha \ \dfrac{1}{\sqrt{GMM}}$

$\therefore r_{O_2} : r_{SO_3} : r_{O_3} = \dfrac{1}{\sqrt{32}} : \dfrac{1}{\sqrt{80}} : \dfrac{1}{\sqrt{48}}$

$= \dfrac{1}{\sqrt{16 \times 2}} : \dfrac{1}{\sqrt{16 \times 5}} : \dfrac{1}{\sqrt{16 \times 3}}$

$= \dfrac{1}{4\sqrt{2}} : \dfrac{1}{4\sqrt{5}} : \dfrac{1}{4\sqrt{3}} = \dfrac{1}{\sqrt{2}} : \dfrac{1}{\sqrt{5}} : \dfrac{1}{\sqrt{3}}$

Hence, the correct option is (c).

15. $2\ NaNO_3/2\ KNO_3 \rightarrow 2NaNO_2/2KNO_2 + O_2$

2 moles of $KNO_3NaNO_3 \rightarrow 1$ mole of O_2

1 mole of $KNO_3NaNO_3 \rightarrow$?

$= 0.5$ moles of $O_2 = 16$ g of O_2 (loss in weight)

$2KClO_3 \rightarrow 2KCl + 3O_2$

2 moles of $KHClO_3 \rightarrow 3$ moles of O_2

1 mole of $KClO_3 \rightarrow$?

$= \dfrac{3}{2} = 1.5$ moles

$= 48$ g of O_2 (loss in weight)

$CaCO_3 \rightarrow CaO + CO_2$

1 mole of $CaCO_3 \rightarrow 1$ mole of CO_2

$= 44$ g of CO_2 (loss in weight)

Hence, the correct option is (b).

CHAPTER 6

Water, Solutions, Solubility and Hydrogen; Metals and Non-metals

Assessment Test I

1. Temporary hardness is imparted to water due to the presence of bicarbonates of calcium and magnesium. Bicarbonates of calcium and magnesium undergo thermal decomposition to form respective carbonates which are insoluble.

   Hence, the correct option is (a).

2. Since carbon atoms are very closely packed in a diamond, it is a good conductor of heat.

   Hence, the correct option is (a).

3. (A) 20 g of solute is present in 180 g of solution B.

   $$\text{Solubility} = \frac{20}{160} \times 100 = 12.5.$$

   (B) 15 g of solute is present in 120 g of solution D.

   $$\text{Solubility} = \frac{15}{105} \times 100 = 14.29.$$

   (C) 30 g of solute is present in 210 g of solution C.

   $$\text{Solubility} = \frac{30}{180} \times 100 = 16.66.$$

   (D) 125 g of solute is present in 625 g of solution A.

   $$\text{Solubility} = \frac{125}{500} \times 100 = 25.$$

   Hence, the correct option is (b).

4. (A) Dressing of ore

   (B) Concentration of ore

   (C) Conversion of ore to oxide

   (D) Reduction of ore

   (E) Refining of ore

   Hence, the correct option is (c).

5.

(i)	Wood charcoal	(C)	Gas masks
(ii)	Activated charcoal	(E)	Adsorbent material
(iii)	Lamp black	(A)	Black shoe polishes
(iv)	Coke	(B)	Extraction of metal
(v)	Animal charcoal	(D)	Removal of colour from sugarcane juice

   Hence, the correct option is (a).

6. Forth floatation method of concentration is inappropriate for an oxide ore.

   Hence, the correct option is (c).

7. Molten slag is obtained at region A.

   Hence, the correct option is (a).

8. One of the products is N_2.

   Hence, the correct option is (a).

9. The given structure indicates white phosphorus. It is not dark red.

   Hence, the correct option is (b).

10. Superheated steam is passed through the outermost pipe to melt the sulphur present below.

    Hence, the correct option is (c).

11. 100 g of solvent can dissolve 20 g of solute.

    ∴ 120 g saturated solution has 20 g of solute.

    ∴ 600 g saturated solution contains $\frac{20}{120} \times 600$ g.

    = 100 g solute

    Hence, the correct option is (a).

12. Molarity of NaOH solution is 5 M.

    Mass of 1l of solution = density × volume

    = (1.5 × 1000) g

    = 1500 g

    Mol. wt. of NaOH = 40

$\therefore$ (40 × 5) g = 200 g of NaOH is present in 1500 g of solution.

$\therefore$ 200 g of NaOH is dissolved in (1500 – 200) = 1300 g of solvent.

Since the solubility of the solute is 20,

1300 g of solvent can dissolve (20 × 13) = 260 g of solute.

Hence, (260 – 200)g = 60 g solute (NaOH) has to be added.

Hence, the correct option is (d).

13. Electrochemical equivalent $= \dfrac{\text{atomic weight}}{\text{valency} \times 96500}$

$\therefore$ Electrochemical equivalent of the metal

$= \dfrac{x}{3 \times 96500}$

Hence, the correct option is (b).

14. $10 \text{ amp} \times (30 \times 60) \text{ s} = 18{,}000$ C electrical charges deposit a g of metal

$\therefore$ 96,500 C electric charges deposits

$\dfrac{a \times 96{,}500}{1800}$ g of metal

Since the metal is divalent, the atomic weight of the metal

$= \dfrac{a \times 96500}{1800} \times 2$

$= \dfrac{a \times 965}{9} = 107.2\, a$

Hence, the correct option is (c).

15. A pale blue tinge in milk occurs due to Tyndall effect because milk is a colloid.

Hence, the correct option is (b).

Assessment Test II

1. Permanent hardness can be removed by adding zeolite. Zeolite replaces Ca^{+2} and Mg^{+2} present in water with Na^{+}.

Hence, the correct option is (c).

2. In graphite, one carbon atom is tightly bonded to three other carbon atoms and the fourth valence electron of the carbon atom loosely binds the two layers of hexagonal structures.

Hence, the correct option is (c).

3. The increasing order of solubilities is ABCDE.

Hence, the correct option is (b).

4. (A) Pulverization

(B) Gravity separation

(C) Roasting

(D) Smelting

(E) Distillation

Hence, the correct option is (d).

5.

(i)	Wood charcoal	(B)	Brittle grey solid
(ii)	Sugar charcoal	(A)	Thick red black substance
(iii)	Coke	(D)	Black porous solid
(iv)	Animal charcoal	(E)	Less percentage of carbon
(v)	Lamp black	(C)	Light powdery black substance having a velvet touch

Hence, the correct option is (d).

6. Cinnabar is a sulphide ore and can be concentrated by froth floatation.

Hence, the correct option is (d).

7. $CaCO_3 \rightarrow CaO + CO_2$

$CO_2 + C \rightarrow CO$

Both the reactions are endothermic, and hence, a drop in temperature is observed in the middle region of the blast furnace.

Hence, the correct option is (d).

8. One of products of the given reaction is NO.

Hence, the correct option is (c).

9. The given structure indicates red phosphorus. Red phosphorus is not used in strike anywhere matches.

Hence, the correct option is (b).

10. Sulphur obtained in Frasch process is in a molten state.

Hence, the correct option is (a).

11. 1 kg saturated solution contains 200 g of solute.

$\therefore$ (1000 – 200) g = 800 g solvent dissolves 200 g of solute.

$\therefore$ 100 g solvent dissolves $\frac{200 \times 100}{800}$ g.

= 25 g of solute

Hence, the correct option is (b).

12. 100 g of solution has 10 g of solute.

$\therefore$ 1 kg of the solution has 100 g of solute.

$\therefore$ (1000 – 100) g = 900 g solvent has 100 g of solute dissolved in it. The solubility of the solute is 25.

$\therefore$ 900 g of solvent can dissolve $\left(\frac{25}{100} \times 900\right)$ g = 225 g of solute.

$\therefore$ (225 – 100) g = 125 g of solute has to be added to saturate the solution.

Hence, the correct option is (c).

13. Electrochemical equivalent = x

$\therefore$ Mass of the metal obtained at the cathode

$= x \times 96{,}500 \times 0.5 = 48{,}250\, x$ g

Hence, the correct option is (b).

14. 1 amp × 1 s = 1 Coulomb

1 Coulomb charge deposits m g of metal.

P Coulombs charge deposits $(P \times m)$ g of metal.

Atomic wt. of the metal is n.

$\Rightarrow$ Valency $= \frac{\text{atomic weight}}{\text{equivalent weight}} = \frac{n}{m \times p}$

Hence, the correct option is (b).

15. Bleeding from wound stops because of coagulation of blood due of the addition of $FeCl_3$ solution. Similarly, fine clay particles which are present in tap water can be coagulated by adding alum to water.

Hence, the correct option is (d).

Assessment Test III

1. Since the specific heat capacity of water is high, it is used as a coolant in car radiators.

Hence, the correct option is (a).

2. Diamond crystals have a very regular arrangement. The forces binding the atoms together are strong and the atoms are close to each other. Hence, they can easily transfer vibratory motion from one atom to the next.

Hence, the correct option is (b).

3. According to Faraday II law, mass ∝ Equivalent weight

$NaCl \Rightarrow E_{Na} = \frac{23}{1} = 23$

$Al_2O_3 \Rightarrow E_{Al} = \frac{27}{3} = 9$

$Fe_2(SO_4)_3 \Rightarrow E_{Fe} = \frac{56}{3} = 18.66$

$ZnCl_2 \Rightarrow E_{Zn} = \frac{65.3}{2} = 32.65.$

Hence, the correct option is (b).

4.

Metal	Ore
Calcium	Gypsum
Iron	Haematite
Lead	Galena
Manganese	Pyrolusite
Thorium	Monazite

Hence, the correct option is (a).

5. Ammonia is used in manufacturing TNT; phosphorus is used in making rat poison; sulphur is used in vulcanization; and chlorine is used in purification of water.

Hence, the correct option is (c).

6. $30 = \frac{W_{solute}}{180} \times 100$

$\Rightarrow W_{solute} = \frac{30 \times 180}{100} = 54$ g

$\therefore W_{solvent} = 180 - 54 = 126$ g

$\therefore V_{water} = \frac{126}{0.9} = 140$ mL

Hence, the correct option is (b).

7.

	25°C	40°C	60°C	
A	110 g	90 g	105 g	$\Rightarrow$ 105 – 90 = 15 g of precipitate
C	130 g	150 g	140 g	$\Rightarrow$150 – 130 = 20 g of precipitate

Hence, the correct option is (d).